

Research Summary

Data-Efficient and Human-Correctable Vision-Language-Action Systems for Real-Robot Manipulation

Research focus: Embodied AI / Vision-Language-Action / Real-robot Learning / Human-in-the-loop Adaptation

Duo Xu | State Key Laboratory of Precision Testing Technology and Instruments, Tianjin University | Email: 2024202134@tju.edu.cn

Research Background

Modern robotic systems need more than visual recognition. To operate reliably in precision manipulation, a robot must convert visual observations into measurable state or error signals, decide what correction is admissible, and verify the outcome under travel, stability, collision, and safety constraints. This sensing-to-action loop is central to visual servoing and also exposes a gap in current vision-language-action (VLA) systems: semantic task understanding and coarse action generation alone do not guarantee accurate, safe physical execution.

My research interest is to connect these two levels. I aim to study VLA systems that provide task-level understanding and coarse decision-making, while vision measurement, constrained local control, and human correction maintain precision, interpretability, and safety. Rather than treating a VLA as a replacement for a verified controller, I view it as one component in a measurable closed-loop system.

Previous Research and Preparation

My master's research focuses on automatic alignment and image-quality optimization for freeform ultra-short-throw projection (USTP) systems. Small six-degree-of-freedom pose errors of optical elements can cause geometric distortion, blur, and MTF degradation. The central problem is therefore to infer useful physical information from camera images and to guide a motion platform toward a better state under real hardware constraints.

I built a high-fidelity non-sequential optical model in Zemax OpticStudio and introduced controlled translations and rotations to study the relationship between pose error and visual degradation. I designed an image-quality evaluation pipeline that uses outer-frame corners and boundary lines for geometric distortion measurement, together with line-pair regions for 93 lp/mm MTF estimation. These measurements turn visual feedback into quantitative objectives for correction.

I then implemented a physics-guided multi-objective adaptive weighting method in C++. During coarse alignment, the optimizer emphasizes geometric distortion; during fine alignment, it increases the weight of MTF. I validated the method on a closed-loop platform consisting of a USTP module, an industrial camera, a six-degree-of-freedom motion platform, an image-quality evaluation program, and a C++ controller. This work was published in Optics Express as "Sensitivity-informed multi-objective adaptive weighting method for automatic alignment of freeform USTP systems."

This experience has trained me to formulate physical tasks as measurable visual error, constrained adjustment, and experimental verification. It also gave me practical preparation in industrial cameras, image processing, C++/Qt, OpenCV, Halcon, Zemax, Python, MATLAB, and hardware-in-the-loop debugging.

Ongoing Embodied-Robotics Preparation

I have now established an SO-101 real-robot preliminary platform with a calibrated leader-follower arm, fixed external and wrist RGB cameras, teleoperation, and hardware-safe rollout procedures. I collected 60 dual-view teleoperated episodes for three language-conditioned tabletop sorting instructions and created episode-level training, validation, and held-out test splits (48/6/6). A fixed-task ACT visual-imitation baseline achieved 8/10 real-robot pick-and-place successes. I also fine-tuned SmolVLA for the three-instruction setting, released the dataset and model configuration, and completed real-robot rollout evaluation. The initial VLA rollout produces target-directed motion, but it does not yet reliably complete the descend-grasp-lift-place sequence under the limited demonstration budget. I retain this result as a documented preliminary failure regime rather than presenting it as a mature multi-task capability. It creates a controlled research question about the roles of visual observability, grasp-stage action completion, initial-state variation, and targeted corrective data.

Future PhD Direction

- Failure-triggered human correction. Compare random new demonstrations, full failed trajectories, and labeled short correction windows under a matched human-time budget.
- Dual-view language-conditioned manipulation. Study how external and wrist observations affect object selection, approach, grasp, lift, and placement failures.
- Constrained action completion. Study when observability and action constraints should stop a policy or request recovery rather than permit unrecoverable motion.

The common research question is: under limited real-world demonstrations, can a hierarchy of VLA-level reasoning, measured visual feedback, safety constraints, and targeted human correction improve robustness and data efficiency without sacrificing interpretability?

Fit with VLA and Precision Manipulation Research

I hope to join a group that studies embodied intelligence together with real robotic systems: visual perception, language-conditioned manipulation, visual servoing, safe learning, or human-robot interaction. My contribution would be an imaging-aware and control-oriented perspective. I am accustomed to identifying dominant error sources, converting visual changes into quantitative measurements, and validating algorithms under calibration drift, mechanical tolerance, and repeated physical trials rather than only on offline datasets.

This transition is focused rather than a change of topic. Optical alignment and precision manipulation share the same methodological structure: measurement, state or error interpretation, constrained adjustment, and verification. In a future high-precision optical-alignment transfer setting, a VLA would provide stage-level suggestions or coarse actions, while a verified vision controller would retain responsibility for fine adjustment, limits, and recovery.

Methodologically, I would bring a full-loop experimental habit to an embodied-AI group. I am used to establishing a measurement protocol before optimizing a policy, isolating the dominant error source, and treating repeated hardware trials as evidence rather than as a final demonstration. This is particularly relevant when a learned action must work with camera calibration, mechanical tolerance, limited viewpoints, and safety constraints.

I also see a constructive division of research responsibilities. VLA research can investigate visual-semantic grounding and generalization; visual measurement can make residual error observable; a safety layer can make action limits explicit; and human feedback can be collected only where it has high value. This structure allows learning-based and model-based components to be compared and improved without confusing a successful video with a verified precision-control result.

During doctoral study, I would first deepen my foundations in robotic perception, language-conditioned manipulation, visual servoing, and safe learning. I would then develop a focused topic around measurement-grounded execution and data-efficient human correction, keeping the real-system constraints visible throughout the research process.

An important benefit of this direction is that it produces a clear evidence chain for a research group: a reproducible real-robot baseline, a language-conditioned policy comparison, measured residual-error and safety analyses, and a human-correction data-efficiency study. Each step has a narrow claim and an identifiable next experiment.

I would be especially motivated by an environment that values this kind of system-level rigor: methods that are ambitious in their learning objective, but also precise about sensing assumptions, operational limits, and the experimental evidence required before transfer to a more demanding physical task.

At the practical level, I would maintain a shared evaluation vocabulary across experiments: the visual cue being measured, the admissible action range, the intervention trigger, the completion rule, and the failure label. This makes it possible to compare a new model with a previous policy without silently changing the task or the safety condition.

My near-term objective is therefore concrete: use the established real-robot evidence base to compare constrained execution and targeted correction strategies before making broader claims about reliable language-conditioned manipulation.

In this sense, the proposed direction is also a training plan for research judgment. It requires deciding which variables are observable, when a controller should defer to a learned policy, and when human input is genuinely more informative than additional data. These are questions that connect model development with the careful use of real hardware.

My first experimental agenda is deliberately compact and falsifiable. I will keep the robot, objects, and task definition fixed while varying one factor at a time: external versus wrist-view availability, object initial position, background or illumination, and the availability of a short human correction. Each rollout will be annotated at the episode level as object-selection, approach, grasp, lift, placement, observability, constraint-stop, or recovery failure. This produces a useful result even when a policy fails: the next data or control intervention is specified by a failure category rather than by intuition alone.

The second comparison will measure data efficiency rather than only final success. Under a matched leader-teleoperation time budget, I will compare additional full demonstrations, complete failed trajectories, and short labeled correction windows. The primary outputs will be success rate with confidence intervals, failure-type distribution, human time, newly collected episodes, and a fixed evaluation protocol. This design makes it possible to distinguish a policy that looks impressive in a selected video from one that improves reliably under controlled initial-state changes.

I would bring to a PhD group a combination of measurement discipline and willingness to report limitations precisely. My optical work has taught me that a closed loop is credible only when the measured signal, allowable adjustment, and validation rule are explicit. The SO-101 study is the first real-robot extension of that habit. It establishes a practical entry point for broader work on embodied learning while keeping claims proportional to the current evidence.